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AI-POWERED DEEP FAKE VIDEO DETECTION USING BI-LSTM FOR STRENGTHENING SOCIAL MEDIA SECURITY

A PROJECT REPORT

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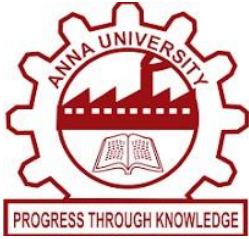
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ABSTRACT

Deepfake technology has emerged as a major threat in the digital era, enabling the creation of highly realistic but manipulated videos that can spread misinformation and compromise digital security. Detecting such fabricated content is a challenging task, as traditional methods often fail to capture subtle spatial and temporal inconsistencies. This project introduces a hybrid deepfake detection framework that integrates Efficient Net for high-accuracy spatial feature extraction, Vision Transformer (ViT) for capturing global contextual relationships and enhancing attention on manipulated regions, and a Bidirectional Long Short-Term Memory (Bi-LSTM) network for temporal sequence modeling. The system supports both video file uploads and YouTube URLs, allowing flexible input options for real-time analysis. Each video is processed and classified as real or fake, with a probability score provided to indicate confidence. A concise report is also generated to summarize the authenticity results. By combining powerful visual representation, global attention mechanisms, and bidirectional temporal modeling, the proposed approach enhances accuracy, robustness, and reliability in identifying manipulated media. This work provides a practical and effective solution to mitigate the growing impact of deepfake threats.

Keywords:

Deepfake Detection, Efficient Net, Vision Transformer (ViT), BiLSTM, Video Authentication, Temporal Analysis, Misinformation, Digital Forensics, Convolutional Neural Networks, Sequence Modeling, Real-time Processing

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
Bi-LSTM	Bidirectional Long Short-Term Memory
GAN	Generative Adversarial Network
DNN	Deep Neural Network
DFT	Deep Fake Technology
DFDC	Deep Fake Detection Challenge
Celeb-DF (v2)	Celeb-DF Version 2 Dataset
SVM	Support Vector Machine
ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
ACC	Accuracy
F1	F1 Score (Harmonic Mean of Precision and Recall)
ViT	Vision Transformer
GPU	Graphics Processing Unit

CHAPTER 1

INTRODUCTION

Deepfake technology has developed rapidly in recent years, making it possible to create highly realistic synthetic videos and images. These manipulated media contents can convincingly change a person's appearance, speech, or actions. Although deepfake technology has useful applications in areas such as entertainment, education, and digital media production, it also introduces serious challenges related to security, privacy, and trust in digital communication.

The misuse of deepfakes can lead to the spread of false information, identity theft, fraud, and damage to personal or organizational reputation. As a result, detecting manipulated videos has become an important research problem in the fields of computer vision and cybersecurity. Traditional deepfake detection techniques mainly rely on handcrafted features or simple visual cues, which are often ineffective in identifying modern deepfakes created using advanced artificial intelligence models such as Generative Adversarial Networks (GANs) and diffusion-based approaches. These fake videos may contain very subtle distortions and micro-expressions that are difficult to detect using conventional static frame analysis methods.

To overcome these challenges, this project proposes a hybrid deepfake detection framework that integrates Efficient Net for extracting detailed spatial features from video frames and Bidirectional Long Short-Term Memory (Bi-LSTM) networks for learning temporal relationships across sequential frames. By combining spatial and temporal analysis, the proposed system improves the accuracy and reliability of deep-fake detection.

The developed system is designed with practical usability in mind. It supports flexible input options such as direct video uploads and video links from online platforms. The model analyses the given video and classifies it as real or fake while providing a probability-based confidence score. In addition, a concise report is generated to help users clearly understand the prediction results.

Overall, this framework aims to provide a scalable and efficient solution to address the growing threat of synthetic media and to enhance the trustworthiness of digital visual content across online platforms.

1.1. SCOPE OF PROJECT

The proposed system is developed to detect Deepfake videos with high accuracy, efficiency, and flexibility. It supports both direct video file uploads and YouTube URL inputs, enabling real-time analysis from diverse content sources. Each processed video is classified as either real or fake, along with a probability score that represents the model's confidence in its prediction.

Additionally, a concise summary report is automatically generated to present the detection results in a clear and interpretable manner.

The framework is scalable, robust, and reliable, making it suitable for practical deployment across multiple domains such as media verification, cybersecurity, digital forensics, law enforcement, and content moderation. By accurately identifying manipulated or forged content, the system helps safeguard individuals and organizations from misinformation, identity theft, and reputational damage.

Moreover, this project establishes a strong foundation for future advancements, including enhanced temporal modeling, multimodal fusion (integrating visual, audio, and textual cues), and real-time monitoring system integration. Through ongoing development and optimization, the proposed framework contributes meaningfully to global efforts aimed at detecting, preventing, and mitigating the threats posed by Deepfake technology.

1.2. OBJECTIVE

The main objective of this project is to design and develop an efficient and reliable deepfake detection framework that leverages the combined strengths of Efficient Net, Vision Transformer (ViT), and Bidirectional Long Short-Term Memory (Bi-LSTM) networks. The

proposed system aims to accurately identify manipulated or synthetic videos by analyzing both spatial and temporal characteristics present in multimedia content.

Efficient Net is utilized for extracting rich and discriminative spatial features from individual video frames. It helps in capturing fine-grained visual details such as texture inconsistencies, lighting variations, and facial artifacts that may indicate possible manipulation. To further enhance feature understanding, the Vision Transformer (ViT) model is incorporated to capture global contextual relationships within frames. ViT improves the model's ability to focus on important regions of the video by applying attention mechanisms, thereby strengthening the detection of subtle forged patterns.

In addition to spatial analysis, Bidirectional Long Short-Term Memory (Bi-LSTM) networks are integrated to model temporal dependencies across sequential frames. This enables the system to identify unnatural motion patterns, frame-level inconsistencies, and sequential anomalies that are commonly present in deep-fake videos. By combining spatial feature extraction with temporal sequence learning, the proposed framework significantly improves the overall detection accuracy and robustness.

Furthermore, the project emphasizes practical usability and real-world applicability. The system supports flexible input options such as direct video file uploads and video links from online platforms like YouTube. It provides probability-based authenticity scores to indicate the confidence level of predictions and automatically generates concise analytical reports for clear interpretation of results.

Overall, the objective of this project is to develop a scalable, accurate, and user-friendly deep-fake detection solution that can assist in combating misinformation, enhancing digital security, and strengthening trust in online visual media.

1.3. MOTIVATION OF THE PROJECT

The motivation for this project arises from the rapid growth and increasing misuse of deepfake

technology in today's digitally connected world. With the advancement of artificial intelligence and deep learning techniques, it has become easier to create highly realistic manipulated videos that can mislead viewers and spread false information. Although deepfake technology offers beneficial applications in areas such as entertainment, virtual media production, and education, its malicious usage in generating fake identities, financial fraud, cyber threats, and reputational damage have raised serious concerns regarding digital trust and security.

Traditional deepfake detection approaches often rely on handcrafted features or simple visual inspection methods, which are insufficient to identify sophisticated forgeries. Modern deepfake videos are generated using advanced generative models that produce temporally consistent and visually convincing content, making manual verification difficult and time-consuming. This creates an urgent need for intelligent and automated detection systems that can analyze complex spatial and temporal patterns in multimedia data.

To address these challenges, the proposed project is motivated by the idea of integrating advanced deep learning architectures for improved detection performance. Efficient Net is utilized for extracting detailed visual features from video frames, while the Vision Transformer (ViT) enhances the system's capability to capture global contextual relationships and focus on manipulated regions through attention mechanisms. In addition, Bidirectional Long Short-Term Memory (Bi-LSTM) networks are incorporated to learn temporal dependencies and sequential inconsistencies across frames.

By combining these powerful models, the project aims to develop a more accurate, robust, and scalable deep-fake detection framework. The ultimate motivation is to contribute toward strengthening cybersecurity, preventing misinformation, supporting digital forensic investigations, and improving the overall reliability of online multimedia content.

1.4. PROBLEM STATEMENT

With the rapid advancement of deep learning and artificial intelligence technologies, the creation of highly realistic deepfake videos has become increasingly easy and accessible.

These manipulated videos can convincingly alter a person's facial expressions, speech, or actions, making it difficult for viewers to distinguish between authentic and fabricated content. As a result, deepfake media poses serious threats such as the spread of misinformation and damage to individual and organizational reputation, financial fraud, cybercrime, and the erosion of public trust in digital platforms.

Detecting deepfake videos is a challenging task because traditional detection methods often rely on basic visual cues or handcrafted features, which are not sufficient to identify modern forgeries.

Advanced deepfake generation techniques produce videos with subtle spatial distortions and temporally consistent motion patterns that are difficult to detect through manual inspection or static frame analysis. Furthermore, the exponential growth of online video content across social media platforms, streaming services, and digital communication channels makes manual verification impractical and time-consuming.

Therefore, there is a strong need for an automated, accurate, and scalable deepfake detection system that can effectively analyze both spatial and temporal characteristics of video data. Such a system should be capable of handling diverse input formats, including direct video uploads and online video links, while providing reliable classification results in real time. Addressing this problem is essential for enhancing cybersecurity, supporting digital forensic investigations, and ensuring the authenticity and trustworthiness of multimedia content in the digital era.

CHAPTER 2

LITERATURE SURVEY

A literature survey is a systematic and critical review of previously published research works related to a specific topic of study. It helps in understanding the existing methodologies, technologies, challenges, and research gaps in the selected problem domain. The main purpose of conducting a literature survey is to gain familiarity with current trends, theoretical concepts, and practical approaches that have been proposed by researchers in the field.

Through the analysis of earlier studies, it becomes possible to identify the strengths and limitations of existing solutions and to justify the need for developing improved or innovative methods. In addition, a literature survey provides a strong foundation for future research by highlighting unexplored areas and emerging challenges. In the context of deepfake detection, reviewing related research enables a better understanding of various detection techniques, datasets, and deep learning models that contribute to the design of an effective and reliable detection framework.

2.1. CNN-BASED DEEPPFAKE DETECTION

Convolutional Neural Networks (CNNs) are widely used in deepfake detection for extracting spatial features from individual video frames. These models analyze important visual characteristics such as facial texture irregularities, blending boundaries, unnatural lighting variations, and pixel-level artifacts introduced during manipulation. By learning hierarchical feature representations, CNN architectures such as Efficient Net and Res Net can effectively distinguish between real and fake facial images.

CNN-based detection systems are particularly useful in identifying frame-level manipulations because they focus on fine visual details that are often difficult to detect through manual observation. They also provide strong generalization capability when trained on large and diverse datasets. However, since CNNs primarily process frames independently, they may fail to capture temporal inconsistencies that occur across

consecutive frames in deepfake videos.

Advantages

- Effectively captures fine-grained spatial and texture-based facial features.
- Provides high classification accuracy for frame-level or image-based deepfake detection.
- Learns strong hierarchical feature representations when trained on large datasets.
- Capable of automatically identifying manipulation artifacts without manual feature engineering.
- Suitable for integration with advanced architectures such as Efficient Net and Res Net.
- Enhance detection performance combined with preprocessing and data augmentation techniques.

Disadvantages

- Unable to model temporal dependencies and motion inconsistencies across sequential frames.
- Detection performance decreases for highly realistic and temporally consistent deepfake videos.
- Deep CNN architectures may require high computational power and longer training time.
- Performance may be affected by variations in lighting conditions, video compression, and image quality.
- Risk of overfitting if trained on limited or biased datasets.
- Requires large, annotated datasets for achieving robust generalization.

2.2. TEMPORAL MODELING USING RNN/LSTM NETWORKS

Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are

widely used in deepfake detection to model temporal relationships present in video sequences. Unlike spatial models that analyze frames independently, these networks process sequential frame information to learn motion patterns, facial expression continuity, and frame transition dynamics. This enables the detection system to identify subtle temporal inconsistencies such as unnatural lip synchronization, irregular blinking patterns, and abrupt facial movements that may indicate manipulation.

Furthermore, Bidirectional LSTM (Bi-LSTM) networks enhance temporal analysis by considering both past and future frame context during feature learning. This bidirectional processing improves the model's ability to understand long-range dependencies and increases detection robustness in complex deepfake scenarios. Temporal modeling methods are therefore essential for improving the reliability of video-based deepfake detection systems.

Advantages

- Effectively captures temporal dependencies and sequential motion patterns.
- Improves detection accuracy in video-based deepfake analysis.
- Help identify subtle inconsistencies such as abnormal facial movements and expression transitions.
- Bidirectional learning enhances contextual understanding of frame sequences.
- Provides better performance when combined with spatial feature extraction models.

Disadvantages

- Requires high computational power and memory for processing long video sequences.
- Training time increases significantly for large-scale datasets.
- May suffer from gradient vanishing or overfitting issues if not properly optimized.
- Performance depends on sequence length selection and frame sampling strategy.
- Complex model tuning and parameter configuration are often required.

2.3. VISION TRANSFORMER-BASED DEEPPFAKE DETECTION

Vision Transformer (ViT) based approaches have recently gained significant attention in deepfake detection due to their ability to capture global contextual relationships in images and videos.

Unlike traditional convolutional models that focus mainly on local spatial features, transformer architectures use self-attention mechanisms to analyze the importance of different regions within a frame. This allows the model to prioritize manipulated facial regions such as eyes, lips, and boundary areas where deepfake artifacts are more likely to occur.

By modeling long-range dependencies, ViT improves the detection of subtle inconsistencies that may not be easily captured by CNN-based methods. These models are particularly useful in identifying complex and high-quality manipulations, thereby enhancing overall detection robustness.

Advantages

- Effectively captures global contextual information and long-range feature dependencies.
- Performs well in detecting small or subtle manipulated regions.
- Improves model interpretability through attention visualization.
- Enhances performance when combined with CNN or temporal models.

Disadvantages

- Requires large-scale annotated datasets for effective training.
- Computationally more expensive compared to traditional CNN architectures.
- Longer training time and higher memory consumption.
- Performance may be reduced in low-quality or highly compressed video inputs.

2.4. FREQUENCY DOMAIN ANALYSIS FOR DEEPPAKE DETECTION

Frequency domain-based detection methods analyze spectral patterns present in images and videos to identify inconsistencies introduced by deep-fake generation models. Synthetic media often fails to replicate the natural frequency distribution of real images, especially in high-frequency regions. By examining noise patterns, texture distribution, and spectral irregularities, these techniques can effectively detect manipulated content.

Such approaches are particularly useful for identifying deep-fakes generated using Generative Adversarial Networks (GANs), as these models tend to introduce characteristic frequency artifacts. Frequency analysis also improves the generalization ability of detection systems across different datasets.

Advantages

- Effective in detecting GAN-generated synthetic images and videos.
- Improves cross-dataset detection performance and generalization.
- Complements spatial deep learning models for better accuracy.
- Helps identify hidden artifacts not visible in pixel space.

Disadvantages

- Sensitive to compression artifacts, resizing, and video noise.
- Performance decreases for highly refined and post-processed deepfakes.
- Requires additional preprocessing steps for frequency transformation.
- May not effectively capture temporal inconsistencies.

2.5. HYBRID CNN–RNN (BILSTM) FRAMEWORK

Hybrid deepfake detection frameworks combine Convolutional Neural Networks (CNNs) with Bidirectional Long Short-Term Memory (Bi-LSTM) networks to perform both spatial and temporal analysis. CNN models extract discriminative visual features from individual frames, while Bi-LSTM networks learn sequential motion patterns and temporal

dependencies across video sequences.

This integrated approach allows the detection system to identify both visual artifacts and unnatural motion transitions present in manipulated videos. As a result, hybrid architecture provides improved robustness and accuracy compared to single-model detection techniques.

Advantages

- Enables simultaneous spatial and temporal feature learning.
- Provides higher detection accuracy and reliability.
- Improves robustness against sophisticated deepfake techniques.
- Suitable for real-world video forensic applications.

Disadvantages

- Increases overall system complexity and model size.
- Requires higher computational resources and training time.
- Difficult hyperparameter tuning and architecture optimization.
- Real-time deployment may be challenging without optimization.

2.6. DATASET-DRIVEN DEEPPFAKE DETECTION APPROACHES

Modern deepfake detection systems rely heavily on large benchmark datasets containing manipulated videos generated under diverse environmental conditions. Training models on such datasets helps improve their generalization capability and performance in real-world scenarios. Dataset-driven approaches allow researchers to evaluate detection accuracy, robustness, and scalability across different manipulation techniques and compression levels. These datasets also support the development of standardized evaluation protocols, enabling fair comparison between various detection methods. However, dataset quality and diversity play a crucial role in determining the effectiveness of trained models.

Advantages

- Facilitates benchmarking and comparative evaluation of detection models

- Improves robustness and generalization in practical applications
- Help models learn diverse manipulation patterns and environmental variations
- Supports large-scale training and performance optimization

Disadvantages

- Dataset bias may negatively affect real-world generalization.
- Requires extensive preprocessing, annotation, and storage resources.
- Training on very large datasets increases computational cost.
- Performance may still degrade when exposed to unseen manipulation techniques.

Table 1. A Summary of the Algorithm, Dataset and Accuracy in Some of the Reviewed Studies

Study (Year)	Algorithm/ Model	Key Features	Dataset(s)	Accuracy / AUC / F1	Notes
Malik et al. (2022)	Survey of DNN-based methods	CNN, RNN, Capsule Networks, Attention	FF++, Celeb- DF, DFDC, etc.	Varies (60–99% AUC)	Comprehensive review; no single model
Tran et al. (2023)	Meta Deepfake Detection (MDD)	Meta- learning, domain-specific weights, pair-attention loss	Multi-domain: FF++, Celeb- DF, DFDC	85.3% AUC (cross-dataset)	Strong generalist of unseen domains
Mubarak et al. (2023)	Survey (Visual + Audio + Text)	CNN, Transformers, Spectrogram analysis	Diverse multimodal datasets	N/A (review)	Focus on impacts and detection trends
Ramadhani et al. (2024)	VIVIT + Facial Landmarks + DSC + CBAM	Landmark tube lets spatial-temporal attention	Celeb-DF v2	87.18% AUC 92.52% F1	Best on challenge low-quality forgeries
Yue et al. (2024)	LFGDI N (GSTN + LRFGM)	Frequency-guided local dynamics (eyes/mouth)	FF++, DFDC, Celeb-DF	91.4% AUC (Celeb-DF), 88.7% (DFDC)	Superior on unknown manipulate types

CHAPTER 3

SYSTEM ANALYSIS

3.1. Existing System

The existing systems for deepfake detection primarily rely on traditional machine learning techniques or simple deep learning models that focus only on spatial features extracted from video frames. Many of these methods analyze still images or individual frames without considering the temporal relationships between consecutive frames, which limits their ability to detect subtle inconsistencies present in manipulated videos. Additionally, some existing approaches depend on handcrafted features such as facial landmarks, head pose estimation, or texture-based analysis, which often fail when deepfakes are generated using advanced generative models that minimize such artifacts. While convolutional neural networks (CNNs) have improved detection accuracy by learning richer visual features, their performance significantly drops when applied to videos with high realism or when tested across different datasets.

Limited to Spatial Analysis: Most existing systems use traditional machine learning or basic CNNs that extract features from individual frames or still images, ignoring temporal dynamics across video sequences.

Dependence on Handcrafted Features: Several approaches rely on manually engineered cues such as facial landmarks, head pose, or texture patterns, which fail against modern generative models that eliminate detectable artifacts.

Lack of Temporal Consistency Modeling: Absence of sequential analysis prevents detection of subtle inter-frame inconsistencies, a key indicator of manipulation in deepfake videos. **Poor Generalization Across Datasets:** Even advanced CNN-based methods suffer significant performance degradation on high-realism videos or when evaluated across diverse datasets due

to domain shift.

3.1.1. Limitations of the Existing System

- **Poor Generalization:** Limited adaptability across diverse datasets and platforms, resulting in reduced accuracy in real-world scenarios.
- **Vulnerability to High-Quality Deepfakes:** Significant performance degradation when encountering realistic, high-fidelity manipulated videos.
- **Low Robustness:** Easily fooled by advanced generative models (e.g., diffusion-based or GAN-enhance) that eliminate detectable visual or temporal artifacts.

3.2. PROPOSED SYSTEM

The proposed system introduces a robust deep-fake detection framework that integrates both spatial and temporal features to overcome the limitations of existing methods. Efficient Net is employed as the backbone model to extract rich and discriminative visual features from individual video frames, ensuring efficient yet powerful spatial representation. To capture global contextual relationships within each frame, a Vision Transformer (ViT) is incorporated, enhancing spatial attention and improving detection of subtle manipulated regions. These features are then processed by a Bidirectional Long Short-Term Memory (Bi-LSTM) network, which captures temporal dependencies and sequential patterns across consecutive frames, allowing the system to detect subtle inconsistencies that often go unnoticed in traditional approaches. Unlike existing systems that focus only on offline datasets or static frames, the proposed system supports flexible input options such as direct video uploads and YouTube URLs, enabling real-world applicability. Each video is classified as real or fake with a probability-based confidence score, and a concise report is generated to summarize the authenticity results. By combining the strengths of Efficient Net, ViT, and BiLSTM, the proposed framework enhances accuracy, scalability, and robustness, offering a practical solution for detecting manipulated media in diverse and dynamic environments.

3.2.1. KEY FEATURES OF THE PROPOSED SYSTEM

a) Hybrid Spatial-Temporal Feature Learning

- Employs Efficient Net for efficient, high-accuracy spatial feature extraction from individual video frames.
- Integrates BiLSTM to model bidirectional temporal dependencies, capturing subtle inconsistencies across frame sequences.
- Utilizes Vision Transformer (ViT) to capture global contextual relationships within each frame, enhancing spatial attention and improving detection of manipulated regions.

b) Flexible Real-World Input Handling

- Supports direct video file uploads and YouTube URL processing for seamless real-time analysis.
- Enables deployment in diverse environments with minimal user friction.

c) Confidence-Aware Deepfake Classification

- Outputs binary classification (Real/Fake) with a probability confidence score for reliable decision-making.
- Enhances interpretability and trustworthiness of detection results.

d) Automated Authenticity Report Generation

- Generates a concise, structured report summarizing classification outcome, confidence level, and key observations.
- Facilitates quick review and documentation for end-users and stakeholders.

e) Scalable and Generalizable Architecture

- Leverages pre-trained Efficient Net for computational efficiency and transfer learning.
- Ensures robust performance across varying video quality, compression levels, and unseen manipulation techniques.

3.2.2. ADVANTAGES OF THE PROPOSED SYSTEM

- Combining Efficient Net for spatial features and Bi-LSTM for temporal patterns improves detection of subtle manipulations.
- Bi-LSTM captures sequential dependencies in video frames, detecting inconsistencies missed by frame-only analysis.
- Supports video file uploads and YouTube URLs, enabling versatile and real-time analysis.
- Provides probability-based confidence scores, enhancing trust and interpretability of detection results.
- Leverages pre-trained Efficient Net for computational efficiency and faster inference without compromising accuracy.

3.3. FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase, and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis some understanding of the major requirements for the system is essential. Three key considerations are examined

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- OPERATIONAL FEASIBILITY

3.3.1.ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the

organization. The amount of funds that the company can pour into the research and development of the system is limited. The expenditure must be justified. Thus, the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only customized products had to be purchased.

3.3.2. TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand for the available technical resources. This will lead to high demands for the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

3.3.3. SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER 4

SYSTEM SPECIFICATIONS

The successful implementation of the deepfake detection system requires appropriate hardware and software resources to ensure efficient video processing, model training, and prediction performance. The system specifications used for the development and testing of the proposed framework are described below.

4.1. HARDWARE REQUIREMENTS

- Processor: Intel Core i5 or higher
- RAM: Minimum 8 GB
- Hard Disk: 500 GB or above
- Graphics: Integrated / Dedicated GPU (optional but recommended for faster deep learning processing).

These hardware components support smooth execution of computational tasks such as video frame extraction, feature processing, and deep learning model inference.

4.2. SOFTWARE REQUIREMENTS

- Operating System: Windows 10 / Windows 11
- Programming Language: Python
- Development Environment: Python IDLE / Anaconda / Visual Studio Code
- Libraries and Frameworks: OpenCV, NumPy, Pandas, TensorFlow / PyTorch, Matplotlib
- Front End: HTML, CSS

The above software tools provide the necessary environment for implementing machine learning algorithms, performing data preprocessing, training deep learning models, and visualizing detection results.

4.3.FUNCTIONAL REQUIREMENTS

Functional requirements describe the main operations and services that the proposed deep-fake detection system must perform. These requirements focus on how the system accepts user input, processes video data, and generates the expected output results.

Input

- Upload of video files in supported formats such as MP4, AVI, or MOV.
- Submission of online video links (for example, video streaming platform URLs).
- User request to start deepfake detection analysis.
- Frame extraction and preprocessing parameters for model processing.

Output

- Classification result indicating whether the given video is Real or Fake.
- Probability-based confidence score representing prediction accuracy.
- Visual or textual report summarizing the detection result.
- Display of processed frames or analysis status on the user interface.

4.4.PYTHON LANGUAGE

Python is a high-level, interpreted programming language widely used for developing machine learning and deep learning applications. It provides simple syntax, extensive libraries, and strong community support, making it suitable for implementing complex artificial intelligence solutions. In this project, Python is used to develop the deepfake detection framework, perform video preprocessing, extract features, and train deep learning models.

Python supports powerful libraries such as OpenCV for image and video processing, NumPy and Pandas for data handling, and TensorFlow or PyTorch for building and training neural network models. These tools enable efficient implementation of spatial and temporal analysis

techniques required for detecting manipulated media.

Additionally, Python allows easy integration with web technologies and user interfaces, which helps in building a practical system that supports video uploads, online video link analysis, and real-time result generation. Its flexibility, scalability, and rich ecosystem make Python an ideal choice for developing intelligent multimedia forensic applications.

4.5. PYTHON LIBRARY

The deepfake detection project requires several Python libraries and deep learning frameworks for video processing, model building, and training. The following libraries were installed using pip commands to ensure compatibility with the project:

a) PyTorch – Version 2.4.1 `pip install torch==2.4.1`

PyTorch is used for building and training deep learning models such as CNN and Bi-LSTM for spatial and temporal feature extraction.

b) Torch vision – Version 0.19.1 `pip install torchvision==0.19.1`

Torchvision provides image and video datasets, pre-trained models, and image transformation utilities for computer vision tasks.

c) TensorFlow – Version 2.17.0 `pip install tensorflow==2.17.0`

TensorFlow is used as an alternative deep learning framework for model development, training, and evaluation, especially for CNN and ViT architecture.

These library installations ensure that the system can perform efficient video preprocessing, deep learning model training, and accurate deepfake detection.

4.6. VISUAL STUDIO CODE

Visual Studio Code (VS Code) is a free, lightweight, and open-source code editor developed by Microsoft. It runs on Windows, macOS, and Linux and provides a versatile environment for coding in multiple programming languages like Python, Java, JavaScript, C/C++, and more. VS Code offers powerful features such as IntelliSense for smart code completion, integrated debugging, Git version control, and a built-in terminal, making coding efficient and streamlined. One of its greatest strengths is its extensibility thousands of extensions allow users to add new languages, themes, linters, and tools, effectively transforming it into a lightweight IDE. Its customizable interface, fast performance, and strong community support make it a popular choice among developers for web development, data science, and general programming tasks.

CHAPTER 5

METHODOLOGY

5.1. DATA COLLECTION & PREPROCESSING

The Data Collection and Preprocessing module is responsible for gathering both real and fake video datasets and preparing them for subsequent processing stages. Videos are decomposed into frames, which are then resized, normalized, and cleaned to remove noise and unwanted distortions. This ensures that the input data remains consistent and suitable for both training and testing the model. Proper preprocessing not only enhances the quality of the data but also improves the efficiency and accuracy of feature extraction.

This module plays a vital role in the deepfake detection system by providing high-quality, standardized input to the learning pipeline. Data is sourced from reliable and publicly available datasets, primarily Celeb-DF v2 and Face Forensics++, which offer a diverse range of authentic and manipulated facial videos. These datasets help the model learn robust distinguishing features necessary for accurate deepfake detection.

After collection, each video is decomposed into individual frames to capture essential visual information. The frames undergo multiple preprocessing operations such as resizing, normalization, and noise reduction, ensuring uniformity in input dimensions. Unwanted distortions, lighting variations, and compression artifacts are minimized during this stage to maintain data integrity.

This standardized preprocessing pipeline ensures that the dataset remains clean, balanced, and optimized for deep learning models. As a result, the preprocessing stage significantly enhances the accuracy, stability, and performance of subsequent processes, including feature extraction and temporal modeling.

The datasets used in this project include Celeb-DF v2 and Face Forensics++, both widely recognized benchmarks for deepfake detection research. Celeb-DF v2 contains 590 real and

5,639 fake videos, featuring high-quality synthetic forgeries of celebrities with minimal visual artifacts. It provides a realistic and challenging dataset for evaluating the robustness of detection models. In contrast, Face Forensics++ consists of 1,000 pristine videos and four distinct facials

manipulation methods Deepfakes, Face Swap, Face2Face, and Neural Textures. This dataset supports multi-method forgery detection, enabling the development of models capable of handling various manipulation techniques. Together, these datasets ensure diversity, authenticity, and complexity, forming a strong foundation for building a generalized and reliable deep-fake detection system.

Table 2. Dataset Comparison Table

Dataset	Key Features	Use Case
Celeb-DF v2	High-quality synthetic forgeries; 590 real + 5,639 fake videos; celebrity-focused	Challenging deepfakes with minimal artifacts
FaceForensics++	1,000 pristine videos; includes 4 manipulation types DeepFakes, FaceSwap, Face2Face, NeuralTextures	Multi-method forgery detection and robustness Evaluation

5.2.FEATURE EXTRACTION USING EFFICIENTNET

The Feature Extraction module employs powerful deep learning architecture, Efficient Net, to extract rich and discriminative features from the preprocessed video frames. Efficient Net is selected for its high accuracy and computational efficiency, achieved through compound scaling of network depth, width, and resolution. This enables the model to learn complex

visual representations with fewer parameters, reducing computational cost without compromising performance.

In this stage, each video frame is passed through the Efficient Net model to capture spatial features such as facial structures, expressions, and subtle texture variations. These features play a crucial role in distinguishing authentic videos from manipulated or synthetic ones. By efficiently learning fine-grained visual details, Efficient Net enhances the model's ability to detect deep fakes even in high-quality and minimally altered videos. Overall, this module ensures the extraction of robust, high-level spatial features that serve as the foundational input for the temporal modeling stage (Bi-LSTM), enabling the system to accurately identify inconsistencies across video frames.

5.3.FEATURE EXTRACTION USING VIT

In this stage, Vision Transformer (ViT) is employed to extract rich spatial features from video frames for deepfake detection. Unlike traditional convolutional neural networks (CNNs) that focus primarily on local patterns, ViT treats each frame as a sequence of image patches and leverages self-attention mechanisms to capture global relationships across the entire frame. This enables the model to detect subtle manipulations that are common in deepfake content. Video frames are first preprocessed by resizing, normalizing, and converting them into tensors compatible with the ViT model. A pretrained ViT model is then used, typically removing the final classification head to obtain high-dimensional feature embeddings. Each frame is passed through this extractor, producing embeddings that encode both local and global spatial information. These frame-level features are then sequentially fed into a temporal model such as a Bidirectional Long Short-Term Memory (Bi-LSTM) network, which captures temporal inconsistencies across frames. The combination of VIT for spatial feature extraction and Bi-LSTM for temporal analysis provides a powerful framework for identifying manipulated videos, offering improved accuracy and generalization over traditional CNN-based approaches.

5.4. TEMPORAL MODELING USING Bi-LSTM

The Temporal Sequence Learning module leverages a Bidirectional Long Short-Term Memory (Bi-LSTM) network to model the temporal dynamics and sequential dependencies across video frames. Unlike unidirectional LSTMs, which process sequences only in the forward direction, Bi-LSTMs propagate information both forward and backward through time. This dual-pass mechanism enables the model to integrate past and future contextual information at each timestep, resulting in richer and more comprehensive representations of dynamic facial behavior. By capturing temporal continuity in expressions, micro-movements, and motion trajectories, the Bi-LSTM effectively identifies subtle temporal inconsistencies such as abrupt changes in head pose, unnatural blink rates, or irregular lip synchronization that often arise during deepfake synthesis but remain imperceptible in isolated frames. This module plays a crucial role in enhancing the system's ability to model temporal coherence, complementing

spatial feature extractors such as Efficient Net to achieve robust discrimination between authentic and manipulated content. The resulting frame-level or sequence-level embeddings generated by the Bi-LSTM are subsequently passed to a final classification layer (e.g., fully connected layers or attention mechanisms) to produce binary or probabilistic predictions indicating the authenticity of the video.

5.5. CLASSIFICATION OF MODULE

The Classification Module serves as the final decision-making stage of the deep-fake detection system. It integrates the spatial features extracted by Efficient Net and Vision Transformer (ViT) along with the temporal features learned by Bi-LSTM, combining them into a unified representation that captures both visual and sequential cues. While Efficient Net provides detailed local feature representations, ViT captures global contextual relationships across each frame, allowing the model to detect subtle manipulations more effectively. This fusion enables the model to leverage complementary information from individual frames and their temporal

dynamics for more robust prediction. The integrated feature vectors are then passed through a series of fully connected (dense) layers, where nonlinear transformations are applied to learn higher-level abstractions. The final output layer employs a SoftMax activation function to classify each input sequence as either real or fake. Additionally, the model produces a probability score representing the confidence level of each prediction, providing interpretability in the classification results. By jointly considering spatial features from Efficient Net and ViT along with temporal characteristics from Bi-LSTM, this module ensures enhanced reliability, accuracy, and generalization capability in distinguishing authentic videos from manipulated ones, effectively synthesizing learned representations to deliver precise and confident deepfake detection outcomes.

5.6. REPORT GENERATION MODULE

The Report Generation module presents the results in a clear and user-friendly manner. It summarizes the classification output, displays whether the video is real or fake, and provides the associated probability score to reflect the system's confidence. The report is designed to be concise and easy to interpret, ensuring that users can quickly understand the authenticity of the analyzed video. This module adds practical value by making the system results accessible to both technical and non-technical users.

5.7. VISUALIZATION

Visualization plays an important role in understanding the performance and behavior of the deep-fake detection system. In this project, visualization techniques are used to clearly present model predictions, performance metrics, and feature behavior.

Graphs such as accuracy and loss curves are used to show the training and validation performance of the model over different epochs. These plots help in identifying issues like overfitting or underfitting during training. Confusion matrices are used to visualize classification results by showing true positives, true negatives, false positives, and false

negatives, which provides a clear understanding of the model's prediction accuracy.

For video-based analysis, frame-level prediction graphs are used to illustrate how the model classifies each frame in a video as real or fake. Attention-based visualization highlights important frames or facial regions that influence the model's decision, helping to interpret the results of the EfficientNet-B7 + Bi-LSTM + Attention architecture.

CHAPTER 6

SYSTEM DESIGN

6.1. SYSTEM ARCHITECTURE

This section outlines the proposed methodology for developing a Deepfake Detection System that integrates Efficient Net, Vision Transformer (ViT), and Bidirectional Long Short-Term Memory (BiLSTM) networks. The system is designed to process video datasets containing both real and manipulated content, extract spatial and temporal features, and classify the authenticity of each video. By combining spatial, temporal, and attention-based modeling, the framework enhances accuracy and robustness in detecting subtle manipulations.

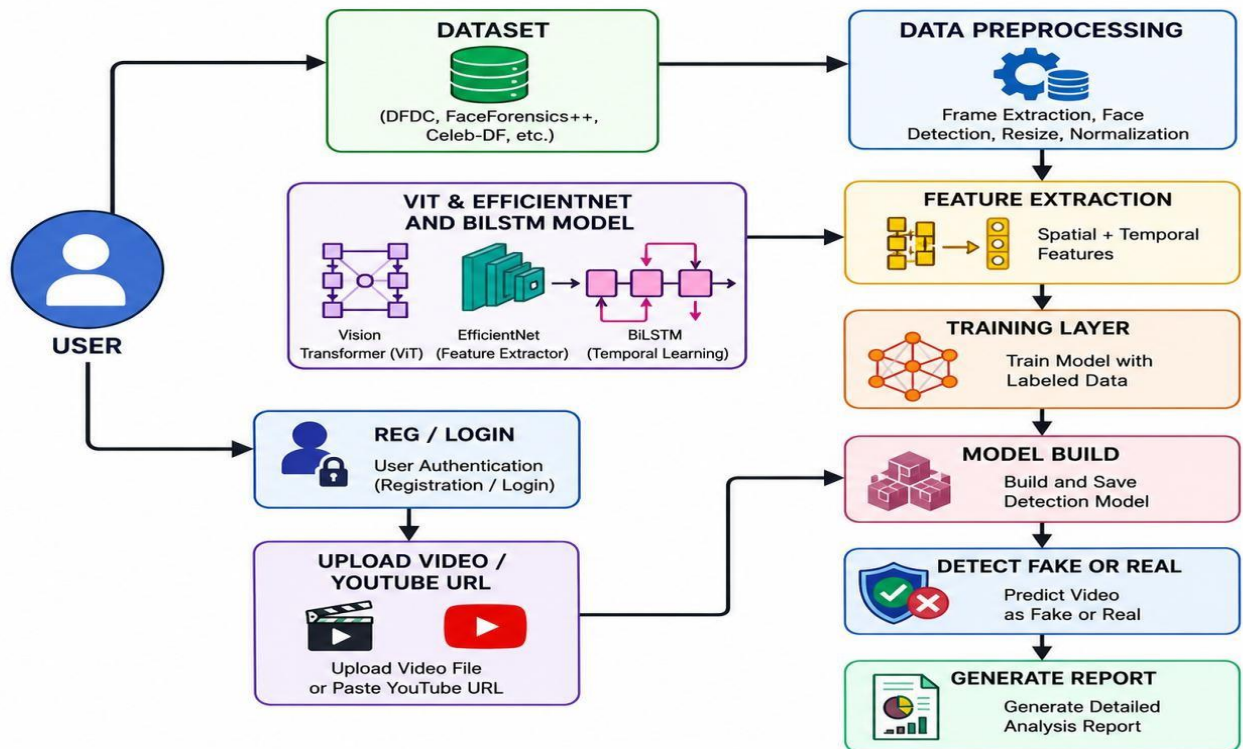


Figure 6.1. System Architecture

6.2.DATA FLOW DIAGRAM

LEVEL 0:

The Level 0 DFD shows the basic flow of data in the Deepfake Detection System.

The system takes Real and Fake Image Datasets as input, performs Data Preprocessing, and then sends the processed data for Feature Extraction.



Figure 6.2. DFD Level 0

LEVEL1:

The Level 1 DFD explains how the Deep Fake Detection System builds its model. It uses VIT, EfficientNet, and Bi-LSTM for feature extraction and temporal modelling to train the model. The built model then takes input videos or URLs for testing and detection.



Figure 6.3. DFD Level 1

LEVEL 2:

The Level 2 DFD represents the final working stage of the Deep Fake Detection System. Here the Web Application allows the user to input a video, which is processed by the trained model to detect whether the video is real or fake.



Figure 6.4. DFD Level 2

6.3. OVERALL DIAGRAM

A Deepfake Detection System is an AI-powered framework designed to automatically identify manipulated or synthetic videos. It analyzes both spatial features from individual video frames and temporal features across sequences of frames to detect subtle inconsistencies caused by manipulation. In this system, Efficient Net extracts detailed local visual features, Vision Transformer (ViT) captures global contextual relationships within each frame, and Bi-LSTM models temporal dependencies between frames. By combining these components, the system classifies videos as either real or fake, providing a probability score for prediction confidence and ensuring accurate, reliable detection of deepfake content.

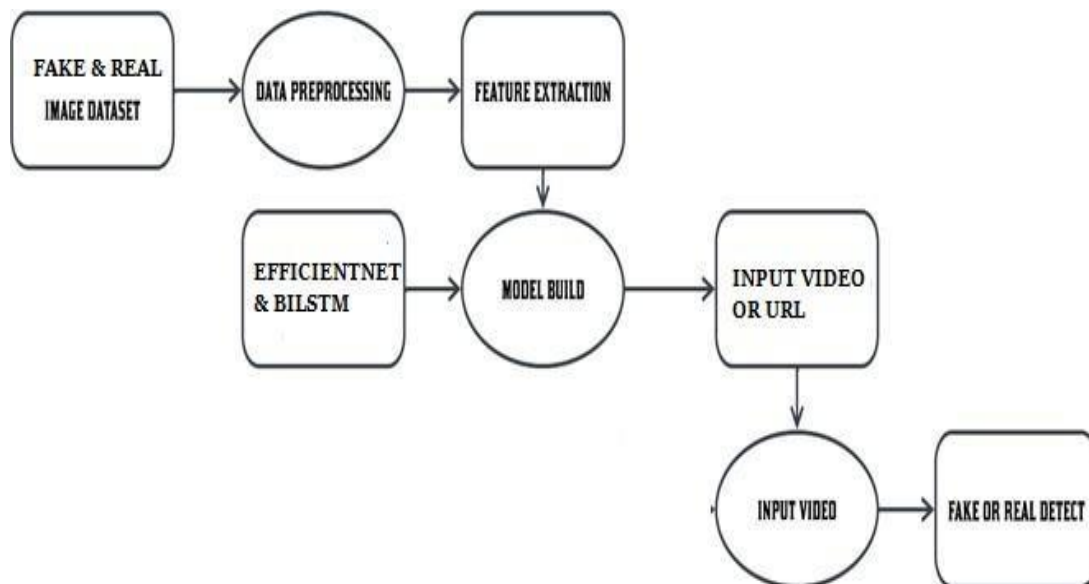


Figure 6.5. Overall Diagram

6.4. USE CASE DIAGRAM

The Deepfake Detection System involves interactions between the user and the system to identify manipulated videos. The user can either upload a local video or provide a URL of an online video for verification. Once the video is submitted, the system first performs preprocessing, which includes extracting frames, resizing, and normalizing them. Next, the system carries out feature extraction, where Efficient Net captures detailed local spatial features and Vision Transformer (ViT) captures global contextual relationships within each frame. These frame-level features are then analyzed temporally using a Bi-LSTM network to detect inconsistencies across consecutive frames. Based on the combined spatial and temporal analysis, the system proceeds to classify the video as either real or fake through fully connected layers and a SoftMax classifier. Finally, the system generates a report presenting the classification result along with a probability score, indicating confidence in the prediction. This use case illustrates a seamless workflow from user input to precise deepfake detection, integrating both spatial and temporal intelligence for robust analysis.

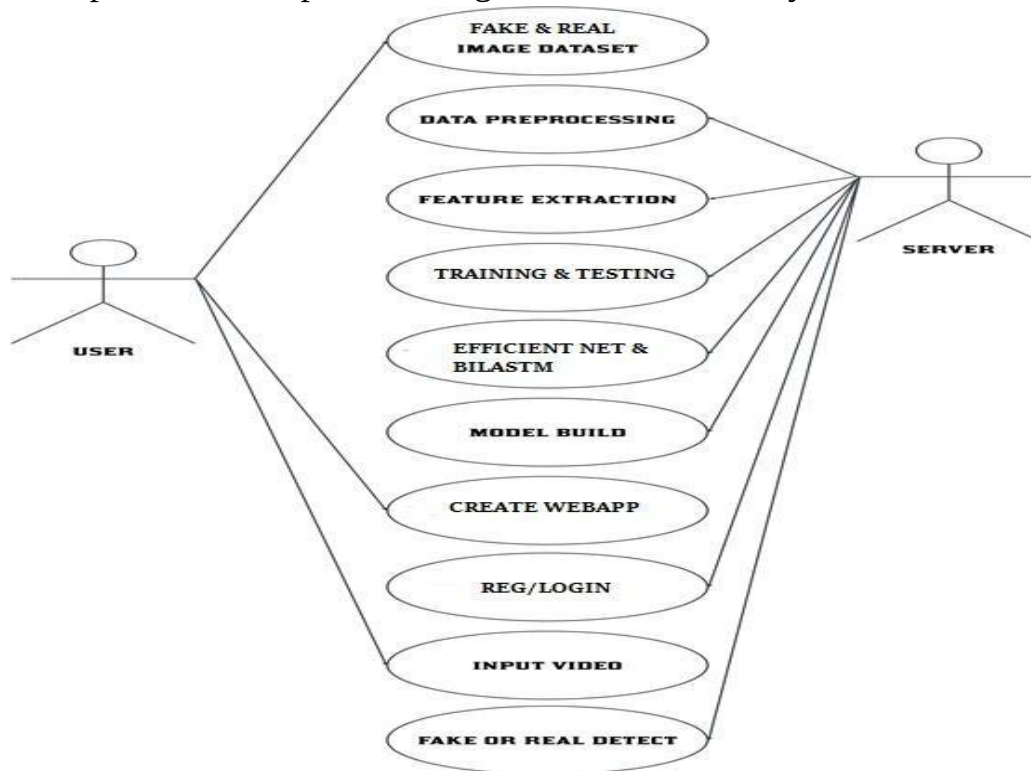


Figure 6.6. Use Case Diagram

6.5. ACTIVITY DIAGRAM

The Activity Diagram of the Deepfake Detection System illustrates the sequential workflow from video input to final classification. The process begins when the user submits a video, either by uploading a local file or providing a URL. The system then performs video preprocessing, which involves extracting individual frames, resizing them, and normalizing pixel values for model compatibility. Each frame is subsequently processed through feature extraction, where Efficient Net captures local spatial details and Vision Transformer (ViT) extract global contextual information. The resulting features from all frames are fused and passed into a Bi-LSTM network, which analyzes temporal relationships and detects inconsistencies across frames. After temporal modeling, the fused features are fed into fully connected layers, where higher-level abstractions are learned, and a SoftMax classifier produces a probability score to classify the video as real or fake. Finally, the system outputs the classification result along with the confidence level, completing the workflow. This activity diagram highlights the logical flow of operations, showing both parallel feature extraction and sequential temporal analysis, ensuring accurate and reliable deepfake detection.

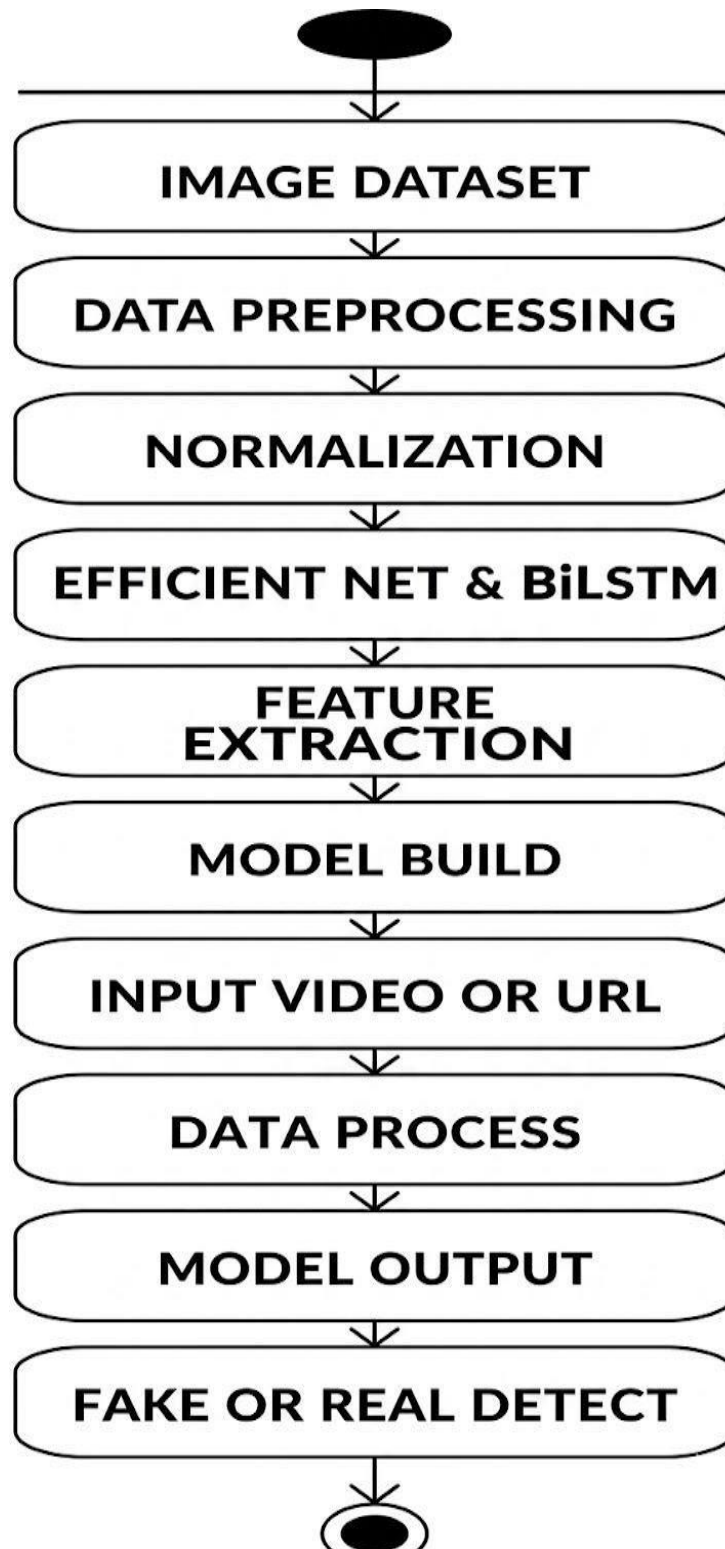


Figure 6.7. Activity Diagram

6.6.LAYERED ARCHITECTURE DIAGRAM

The Overall System Diagram of the Deepfake Detection System presents a comprehensive view of the workflow from video input to final classification. The process begins with the user providing a video, either by uploading a local file or submitting an online video URL. The system then performs preprocessing, which includes extracting individual frames, resizing, and normalizing them. Each frame is subsequently processed through the Feature Extraction Module, where Efficient Net captures local spatial features and Vision Transformer (ViT) capture global contextual relationships, generating rich frame-level embeddings. These features are fused and passed to the Bi-LSTM network, which models temporal dependencies across frames to detect subtle inconsistencies. The Classification Module then receives the fused spatial-temporal features, passes them through fully connected layers, and applies a SoftMax classifier to determine whether the video is real or fake, along with a probability score indicating confidence. The diagram emphasizes the flow of data through parallel spatial feature extraction and sequential temporal modeling, culminating in a precise and interpretable deepfake detection output.

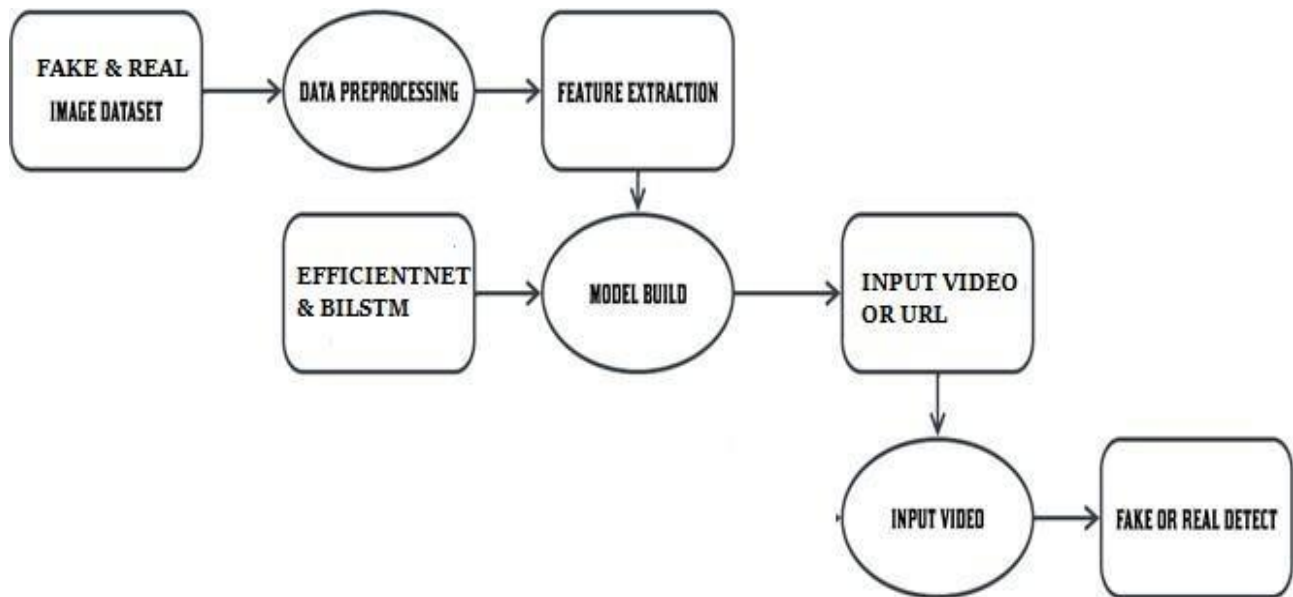


Figure 6.8. Layered Architecture Diagram

CHAPTER 7

PROPOSED SYSTEM ALGORITHM

7.1. EFFICIENTNET MODEL

In the field of deep learning, the quest for more efficient neural network architecture has been ongoing. Efficient Net has emerged as a beacon of innovation, offering a holistic solution that balances model complexity with computational efficiency. This article embarks on a detailed journey through the intricate layers of Efficient Net, illuminating its architecture, design philosophy, training methodologies, performance benchmarks, and more.

7.1.1.EFFICIENTNET

Efficient Net is a family of convolutional neural networks (CNNs) that aims to achieve high performance with fewer computational resources compared to previous architectures. It was introduced by Mingling Tan and Quoc V. Le from Google Research in their 2019 paper "Efficient Net: Rethinking Model Scaling for Convolutional Neural Networks." The core idea behind Efficient Net is a new scaling method that uniformly scales all dimensions of depth, width, and resolution using a compound coefficient.

7.1.2.EFFICIENTNET-B0 ARCHITECTURE OVERVIEW

The EfficientNet-B0 network consists of:

a) Stem

- Initial layer with a standard convolution followed by a batch normalization and a ReLU6 activation.
- Convolution with 32 filters, kernel size 3x3, stride 2.

b)Body

- Consists of a series of MBConv blocks with different configurations.
- Each block includes depth wise separable convolutions and squeeze-and-excitation layers.
- Example configuration for MBConv block:

- Expansion ratio: The factor by which the input channels are expanded.
- Kernel size: Size of the convolutional filter.
- Stride: The stride length for convolution.
- SE ratio: Ratio for squeeze-and-excitation.

c)Head

- Includes a final convolutional block, followed by a global average pooling layer.
- A fully connected layer with a SoftMax activation function for classification.

7.1.3.EfficientNet-B0 Detailed Architecture

Efficient Net uses a technique called compound coefficient to scale up models in a simple but effective manner. Instead of randomly scaling up width, depth, or resolution, compound scaling uniformly scales each dimension with a certain fixed set of scaling coefficients. Using this scaling method and Auto ML, the authors of Efficient Net developed seven models of various dimensions, which surpassed the state-of-the-art accuracy of most convolutional neural networks, and with much better efficiency.

At the heart of Efficient Net lies a revolutionary compound scaling method, which orchestrates the simultaneous adjustment of network width, depth, and resolution using a set of fixed scaling coefficients. This approach ensures that the model adapts seamlessly to varying computational constraints while preserving its performance across different scales and tasks.

7.1.4.Compound Scaling

The authors thoroughly investigated the effects that every scaling strategy has on the effectiveness and performance of the model before creating the compound scaling method. They concluded that, although scaling a single dimension can help improve model performance, the best way to increase model performance overall is to balance the scale in all three dimensions (width, depth, and image resolution) while taking the changeable available resources into consideration.

The below images show the different methods of scaling:

1. **Baseline:** The original network without scaling.
2. **Width Scaling:** Increasing the number of channels in each layer.
3. **Depth Scaling:** Increasing the number of layers.
4. **Resolution Scaling:** Increasing the input image resolution.

Compound Scaling: Simultaneously increasing width, depth, and resolution according to the compound scaling formula.

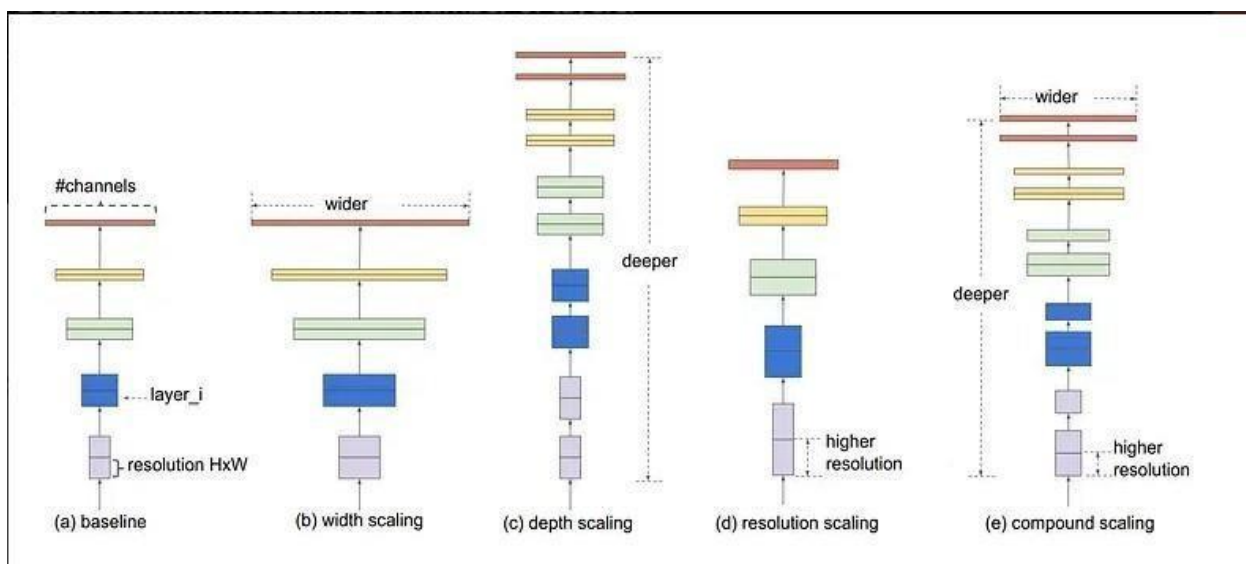


Figure 7.1. EFFICIENTNET-B0 DETAILED ARCHITECTURE

7.1.5. Efficient Scaling

Efficient Net achieves efficient scaling by progressively increasing model depth, width, and resolution based on the compound scaling coefficient ϕ . This allows for the creation of larger and more powerful models without significantly increasing computational overhead. By carefully balancing these dimensions, Efficient Net achieves state-of-the-art performance while remaining computationally efficient. Efficient Attention Mechanism:

Efficient Net incorporates efficient attention mechanisms, such as squeeze-and-excitation (SE) blocks, to improve feature representation. SE blocks selectively amplify informative features by learning channel-wise attention weights. This enhances the discriminative power of the

network while minimizing computational overhead.

7.1.6. Variants of Efficient Net Model:

Efficient Net offers several variants, denoted by scaling coefficients like B0, B1, B2, etc. These variants differ in depth, width, and resolution based on the compound scaling approach.

- **EfficientNet-B0:** The baseline model with moderate depth, width, and resolution.
- **EfficientNet-B1 to B7:** Successively larger variants achieved by increasing the compound scaling coefficient ϕ .
- **Efficient Net-Lite:** Lightweight variants designed for mobile and edge devices, achieving a good balance between performance and efficiency.

7.2. VISION TRANSFORMER

The Vision Transformer (ViT) is a deep learning model that applies Transformer architecture to image classification tasks. Instead of using convolution operations, ViT processes an image as a sequence of fixed-size patches and learns visual representations using a self-attention mechanism. The input image is divided into non-overlapping patches, and each patch is flattened and converted into an embedding vector. Positional embeddings are added to retain spatial information.

These embeddings are passed through multiple Transformer encoder layers, each consisting of multi-head self-attention, feed-forward neural networks, residual connections, and layer normalization. The self-attention mechanism enables the model to capture global dependencies between different regions of the image, making it effective for recognizing complex visual patterns. A special classification token is used to represent the entire image, and its final representation is passed to a classification head to generate the output. Overall, ViT provides a powerful alternative to convolution-based models by effectively modeling long-range relationships in images.

are jointly considered, significantly improving the accuracy and reliability of the deepfake detection system.

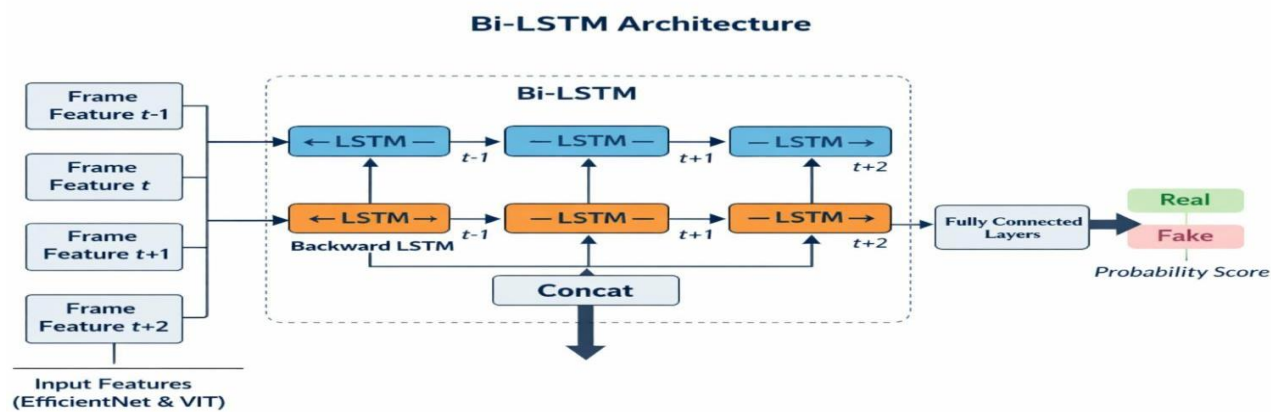


Figure 7.3. Bidirectional Long Short-Term Memory

CHAPTER 8

IMPLEMENTATION

8.1. CODING

home.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Deepfake Detection</title>
<link rel="icon" type="image/png" href="{{ url_for('static', filename='icon/title.png') }}">

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;500;700&display=swap"
rel="stylesheet">

<style>
* {
margin: 0;
padding: 0;
box-sizing: border-box;
font-family: 'Poppins', sans-serif;
}

body {
height:100vh; display: flex;

justify-content: center; align-items: center;
```

```

background: radial-gradient (circle at top left, #051937, #0c2d57, #1b4965, #243b55);
overflow: hidden;
}

.container {display: flex;

align-items: center;
justify-content: space-between; width: 85%;

max-width:1200px; color: #fff;

}

.image-section
{flex:1; display: flex;

justify-content: center; align-items: center;

}

.image-section      img {width: 85%;

max-width: 450px;
filter: drop-shadow(0 20px #00f0ff); animation: float 4s ease-in-out infinite;

}

@keyframes float {
0%, 100% {transform: translateY(0px);} 50% {transform: translateY(-10px);}

}

.text-section {flex: 1;

padding-left: 40px;
}

```

```

.text-section h1 {font-size: 2.5rem; margin-bottom: 15px; letter-spacing: 1px; color:
#00f0ff;
}
.text-section p {line-height: 1.7; font-size: 1rem; color: #d9e6f2;
margin-bottom: 25px;
}

.text-section button {background: transparent; border: 2px solid #00f0ff; color: #00f0ff;
padding: 10px 25px; font-size: 1rem; border-radius: 25px; cursor: pointer; transition:
all 0.3s ease;
}
.text-section button: hover {background: #00f0ff; color: #0c2d57;
box-shadow: 0 0 15px #00f0ff;
}

. popup-form {position: fixed; top: 0;
left: 0;
width: 100%;
height: 100%;
background: rgba (0, 0, 0, 0.8); display: none;
justify-content: center; align-items: center;
}

. form-box {
background: #0c2d57; padding: 30px;
border-radius:15px; width: 350px;

```

```

color: #fff;
box-shadow: 0 0 25px #00f0ff; text-align: center;
}
.form-box h2 {color: #00f0ff; margin-bottom: 20px;
}

.form-box input {width: 100%; padding: 10px;
margin: 10px 0; border-radius: 8px; border: none; outline: none; background: #1b4965; color:
#fff;
}
.form-box button {background: #00f0ff; color: #0c2d57;
border: none; padding: 10px 20px; border-radius: 25px; cursor: pointer; transition: all 0.3s
ease; margin-top: 10px;
}
.form-box button: hover {background: #05a4a6; color: #fff;
}

.close-btn {
background: transparent; border: none;
color: #fff;
font-size: 20px; position: absolute; top: 20px;
right: 30px; cursor: pointer;
}
@media (max-width: 900px) {
.container {
flex-direction: column; text-align: center;

```

```

}
.text-section {padding-left: 0; margin-top: 20px;
}
.image-section img {width: 70%;
}
}
</style>
</head>
<body>
<div class="container">
<div class="image-section">

</div>
<div class="text-section">
<h1>DEEPFAKE DETECTION</h1>
<p>
Welcome to our deepfake detection module. Our advanced AI algorithms analyze videos and
images for signs of manipulation or tampering. Our solution provides a reliable and efficient
way to detect and prevent digital forgery. Upload your media, and our system will generate a
detailed report highlighting any potential anomalies.
</p>
<button id="openForm">Get Started</button>
</div>
</div>
<div class="popup-form" id="popupForm">
<div class="form-box">
<button class="close-btn" id="closeForm">&times;</button>
<h2>Register</h2>

```

```

<form action="/register" method="POST">
<input type="text" name="username" placeholder="Username" required>
<input type="email" name="email" placeholder="Email" required>
<input type="password" name="password" placeholder="Password" required>
<button type="submit">Register</button>
</form>

<h2 style="margin-top:20px;">Login</h2>
<form action="/login" method="POST">
<input type="email" name="login_email" placeholder="Email" required>
<input type="password" name="login_password" placeholder="Password" required>
<button type="submit">Login</button>
</form>
</div>
</div>

<script>
Conspopup=document.getElementById('popupForm');
document.getElementById('openForm').onclick = () => popup.style.display = 'flex';
document.getElementById('closeForm').onclick = () => popup.style.display = 'none';
window.onclick = (e) => { if (e.target == popup) popup.style.display = 'none'; };

</script>
</body>
</html>

```

Upload.html

```

<!DOCTYPE html>
<html lang="en">
<head>

```

```

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Deepfake Detection</title>
<link rel="icon" type="image/png" href="{{ url_for('static', filename='icon/title.png') }}">

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;500;700&display=swap"
rel="stylesheet">
<style>
* {
margin: 0;
padding: 0;
box-sizing: border-box;
font-family: 'Poppins', sans-serif;
}

body {
height:100vh; overflow: hidden;
}

/* Background Video */ video#bg-video{ position: fixed;
top: 0;
left: 0;
width: 100%;
height: 100%; object-fit: cover; z-index: -1;
filter: brightness (0.4);
}

/* Center Container */
.upload-container {position: absolute; top: 50%;
left: 50%;
transform: translate (-50%, -50%); width: 420px;

```

```

background: rgba(10, 25, 47, 0.8); border-radius: 20px;

padding: 40px 30px;
box-shadow: 0 0 25px #00f0ff; text-align: center;

color: #fff;
backdrop-filter: blur(8px);
}

.upload-container h1 {color: #00f0ff; font-size: 1.8rem; margin-bottom: 15px; letter-spacing:
1px;
}

.upload-container p {font-size: 0.95rem; color: #d9e6f2; margin-bottom: 25px;
}

.upload-form {display: flex; flex-direction: column; align-items: center;
}

input[type="file"], input[type="text"] {

width: 100%;
background: #112240; border: none;

outline: none;
padding: 12px; border-radius: 8px; color: #fff;

margin-bottom: 20px; font-size: 0.95rem;

box-shadow: inset 0 0 5px rgba (0, 240, 255, 0.5);
}

```

```

input [type="file"]::file-selector-button {background: #00f0ff; border: none;
color: #0c2d57; font-weight:600; border-radius: 6px; padding: 8px 12px; cursor: pointer;
transition: all 0.3s ease;
}
input [type="file"]:: file-selector-button: hover {background: #05a4a6; color: #fff;
}
button {
background: transparent;
border: 2px solid #00f0ff; color: #00f0ff;
padding: 10px 25px; border-radius: 25px; cursor: pointer;
font-size:1rem; font-weight: 500;
transition: all 0.3s ease;
}

button: hover {background: #00f0ff; color: #0c2d57; box-shadow: 0 0 15px #00f0ff;
}
@media (max-width: 500px) {
.upload-container {width: 90%; padding: 30px 20px;
}
}
</style>
</head>
<body>
<!-- Background video -->
<video autoplay muted loop id="bg-video">
<source src="{ {url_for('static', filename='background video/background.mp4')}}"

```

```

type="video/mp4">

</video>
<!-- Upload Form Container -->
<div class="upload-container">
<h1>Deepfake Detection</h1>
<p>Upload a video file or provide a video URL for deepfake analysis. </p>
<form class="upload-form" action="/detect" method="POST" enctype="multipart/form-
data">
<input type="file" name="video_file" accept="video/*">
<input type="text" name="video_url" placeholder="Enter Video URL">
<button type="submit">Detect Deepfake</button>
</form>
</div>
</body>
</html>

```

result.html

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Deepfake Result</title>
<link rel="icon" type="image/png" href="{{url_for('static', filename='icon/title.png')}}">
<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;600&display=swap"
rel="stylesheet">
<style> body {

```

```
background: radial-gradient (circle at center, #0a192f, #020c1b); color: white;
font-family: 'Poppins', sans-serif;
display: flex;
flex-direction: column; justify-content: center; align-items: center; height: 100vh;
text-align: center; padding: 20px;
}
```

```
. result-box {
border: 2px solid {{color}}; padding: 30px;

border-radius: 20px;
box-shadow: 0 0 25px {{color}}; max-width: 500px;

background: rgba (10,25,47,0.9); animation: glow 1.5s infinite alternate;
}
```

```
@keyframes glow {
from {box-shadow: 0 0 15px {{color}};} to {box-shadow: 0 0 35px {{color}};}
}
```

```
h1 {
font-size: 2rem; color: {{color}};

text-shadow: 0 0 15px {{color}}; margin-bottom: 15px;
}

p {
margin-top: 10px; font-size: 1.1rem; line-height: 1.5;
}
```

```

a {
margin-top: 25px; display:inline-block; padding: 12px 25px;

border: 1px solid #00f0ff;
border-radius: 8px; color: #00f0ff;

text-decoration: none; transition: all 0.3s ease;

}
a: hover {
background:#00f0ff; color: #0a192f;

box-shadow: 0 0 15px #00f0ff;
}
</style>
</head>
<body>
<div class="result-box">
<h1>Prediction: {{result}} </h1>

```

```

{% if result == "REAL" %}
<p>This video appears to be authentic and does not show signs of manipulation.
Confidence in this assessment: <strong> {{confidence}} </strong>. </p>
{% else %}
<p>This video may be altered or artificially generated. It is likely a deepfake.
Confidence in this assessment: <strong> {{confidence}} </strong>. </p>
{% endif %}

<a href="{{url_for('upload_page')}}" ">Back to Upload</a>
</div>

```

</body>

</html>

app.py

```
from flask import Flask, render_template, request, redirect, flash, session import torch
```

```
from torch import nn
```

```
from torchvision import models, transforms from torch.utils.data
```

```
import Dataset
```

```
import cv2
```

```
import face_recognition import numpy as np
```

```
from werkzeug. utils
```

```
import secure_filename import requests
```

```
from urllib. parse import urlparse import os
```

```
import sqlite3 import yt_dlp
```

```
app = Flask(____name____) app. secret_key = "7103"
```

```
app.config['UPLOAD_FOLDER'] = 'Uploaded_Files'
```

```
os. mkdirs(app.config['UPLOAD_FOLDER'], exist_ok=True)
```

```
DB_FILE = "users.db"
```

```
MODEL_PATH = "model/df_model.pt"
```

```
def init_db ():
```

```
with sqlite3.connect(DB_FILE) as conn: conn. execute ("""
```

```
CREATE TABLE IF NOT EXISTS users (
```

```
id INTEGER PRIMARY KEY AUTOINCREMENT,
```

username TEXT NOT NULL, email TEXT UNIQUE NOT NULL,

password TEXT NOT NULL

) """)

init_db ()

class Model (nn. Module):

def __init__(self, num_classes, latent_dim=2048, lstm_layers=1, hidden_dim=2048,
bidirectional=False):

super (Model, self)._init_ ()

model=models. resnext50_32x4d(pretrained=True) self. model = nn. Sequential (*list (model.
Children ()) [: -2])

self. lstm = nn. LSTM (latent_dim, hidden_dim, lstm_layers, bidirectional)

self. dp= nn. Dropout (0.4)

self. linear1=nn. Linear (2048, num_classes) self. avgpool = nn. AdaptiveAvgPool2d (1)

def forward (self, x):

batch_size, seq_length, c, h, w = x. shape

x = x. view (batch_size * seq_length, c, h, w)

fmap= self. model(x)

x = self. avgpool(fmap)

x = x. view (batch_size, seq_length, 2048) x_lstm, _ = self. lstm (x, None)

return fmap, self. dp (self. linear1(x_lstm[:, -1, :]))

im_size = 112

mean = [0.485, 0.456, 0.406]

std = [0.229, 0.224, 0.225]

sm = nn. Softmax ()

```

class ValidationDataset (Dataset):
def __init__ (self, video_names, sequence_length=20, transform=None): self. Video_names =
video_names

self. Transform=transform self. Count = sequence_length

def __len__(self):
return len (self. Video_names)

def __getitem__(self, idx):
video_path = self. Video_names[idx] frames = []

for frame in self. frame_extract(video_path): faces = face_recognition. face_locations(frame)
try: top, right, bottom, left = faces [0]
frame = frame [top: bottom, left: right,:] except:
pass frames. append (self. Transform(frame)) if len(frames) == self. Count:

break

frames = torch. Stack(frames) frames = frames [: self. Count] return frames. unsqueeze (0)

def frame_extract (self, path): vidObj
=cv2.VideoCapture(path) success = True

while success:
success, image = vidObj. Read() if success:

yield image

def predict (model, img): fmap, logits = model(img) logits = sm(logits)

_, prediction = torch.max (logits, 1)
confidence = logits [ :, int (prediction. Item ())]. item () * 100 return [int (prediction. Item ()),

```

confidence]

```
def detect_fake_video(video_path): train_transforms = transforms.Compose([
transforms.ToPILImage (),
transforms.Resize ((im_size, im_size)),
transforms.ToTensor (), transforms.Normalize (mean, std)
])
video_dataset = ValidationDataset([video_path], sequence_length=20,
transform=train_transforms)

model = Model (2)
model.load_state_dict (torch. Load (MODEL_PATH, map_location=torch.
Device('cpu')))) model.eval ()

prediction = predict (model, video_dataset [0]) return prediction

@app.route('/') def home():
return render_template('home.html')
@app.route('/upload') def upload_page():
if 'user' not in session:
flash("Please log in to access this page.", "error") return redirect('/')
return render_template('upload.html')
@app.route ('/detect', methods=['POST']) def detect ():
if 'user' not in session:
flash ("Please log in to use this feature.", "error") return redirect ('/')

temp_video_path = None
if 'video_file' in request. Files and request. Files['video_file']. filename!= "": file
= request. Files['video_file']
filename = secure_filename (file. Filename)
```

```

temp_video_path = os. path. Join (app.config['UPLOAD_FOLDER'], filename)

file. Save(temp_video_path)

elif 'video_url' in request. Form and request. Form['video_url']: video_url = request.
Form['video_url']

filename = "downloaded_video.mp4"
temp_video_path = os. path. Join(app.config['UPLOAD_FOLDER'], filename)
try:
if video_url. Lower (). endswith(('mp4','avi','mov','mkv')): response = requests. Get
(video_url, stream=True, timeout=10) if response. status_code == 200:

with open (temp_video_path, 'wb') as f:
for chunk in response. iter_content(chunk_size=1024): if chunk:

f. write(chunk)
else:
flash ("Failed to download direct video file.", "error") return redirect('/upload')

else:
ydl_opts = {
'outtmpl': temp_video_path, 'quiet': True, 'format': 'best[ext=mp4]/best'
}
With yt_dlp. YoutubeDL(ydl_opts) as ydl:
ydl. download([video_url])
print ("Video downloaded successfully:", temp_video_path) except Exception as e:
flash (f"Error downloading video: {str(e)}", "error") return redirect('/upload')

try:
print ("Video Path:", temp_video_path)
print      ("File      size:",      os.path.getsize(temp_video_path))      prediction      =
detect_fake_video(temp_video_path) output = "REAL" if prediction [0] == 1 else "FAKE"

```

```

color = "lime" if output == "REAL" else "red" confidence = f"{prediction[1]:.2f} %"

return render_template ("result.html", result=output, confidence=confidence, color=color)
except Exception as e:

flash (f"Prediction failed: {str(e)}", "error") return redirect('/upload')

finally:
if temp_video_path and os. path. Exists(temp_video_path): os. Remove(temp_video_path)

@app. Route ('/register', methods=['POST']) def register ():

username = request. Form['username'] email = request. Form['email'] password = request.
Form['password'] try:

with sqlite3.connect(DB_FILE) as conn:
conn. execute ("INSERT INTO users (username, email, password) VALUES (?, ?, ?)",
(username, email, password))
flash ("Registration successful! Please login.", "success") except sqlite3.IntegrityError:

flash (Email already Exists!,"error") return redirect ('/')

@app. route ('/login', methods=['POST']) def login ():

email =request. Form['login_email'] password = request. Form['login_password']

with sqlite3.connect(DB_FILE) as conn: cur = conn.cursor()

cur. executes ("SELECT * FROM users WHERE email =? AND password =?", (email,
password))

user = cur. fetchone ()

if user:
session['user'] = user [1] flash (f"Welcome, {user[1]}!", "success") return redirect('/upload')

else:
flash ("Invalid credentials!", "error") return redirect ('/') @app. route('/logout')

```

```
def logout (): session. Pop ('user', None)
```

```
flash ("Logged out successfully.", "success") return redirect ('/') if __name__ == '__main__':
```

```
app.run (debug=False, port=7103)
```

8.2. OUTPUT

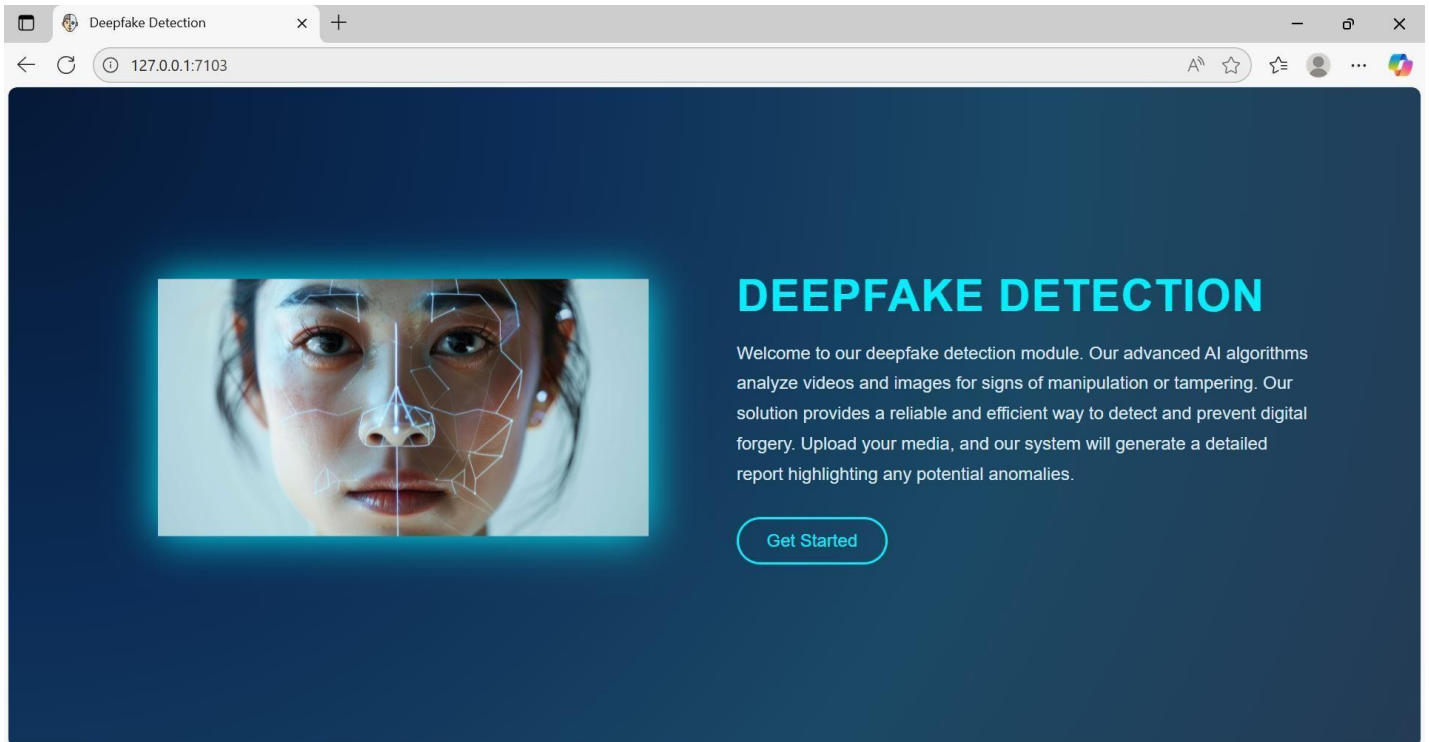


Figure 8.1. Deepfake Detection Home Page

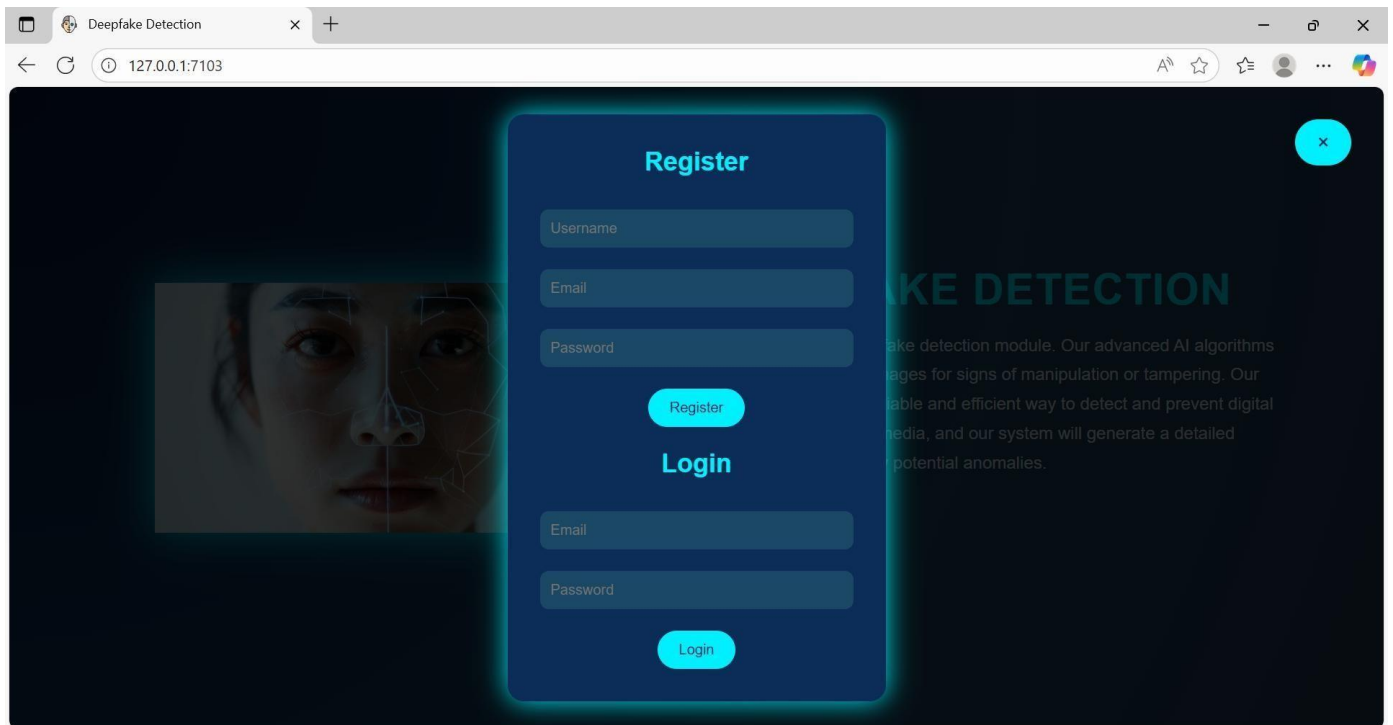


Figure 8.2. Deepfake Detection Authentication Pages

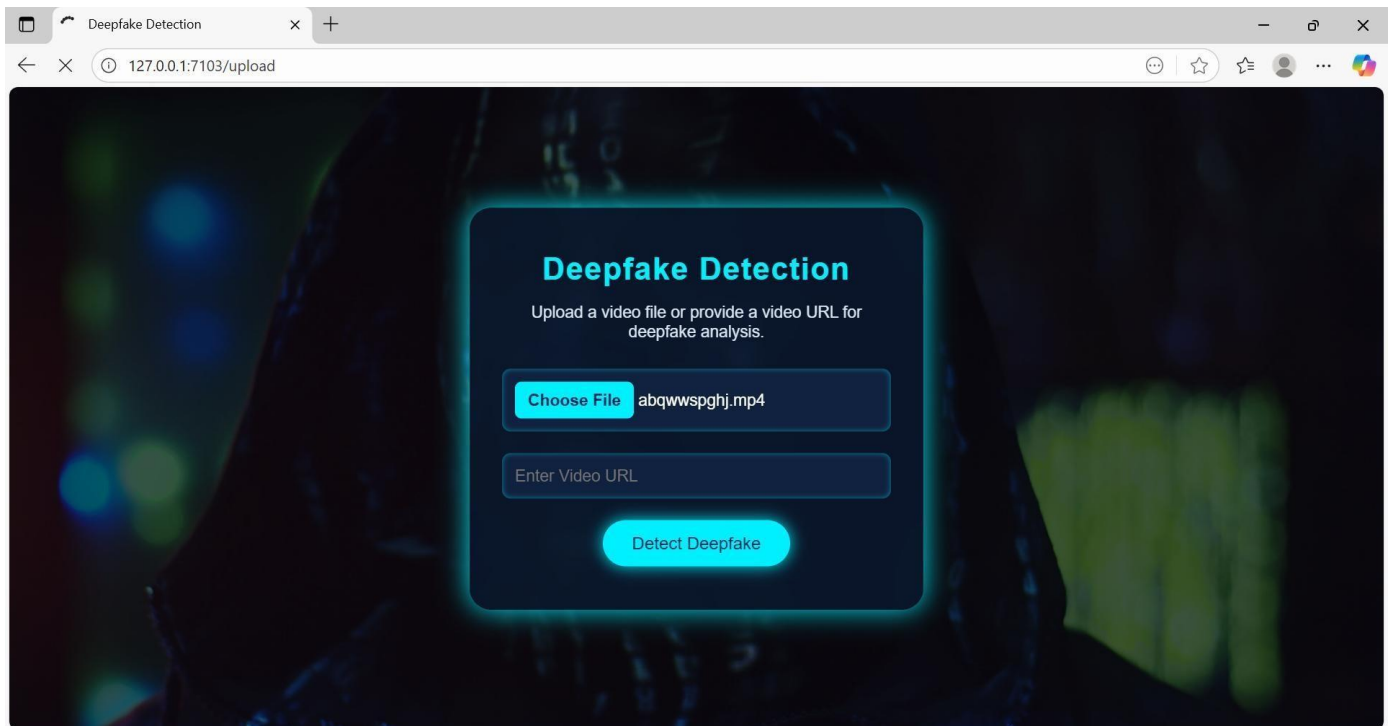


Figure 8.3. Upload & YouTube Detection

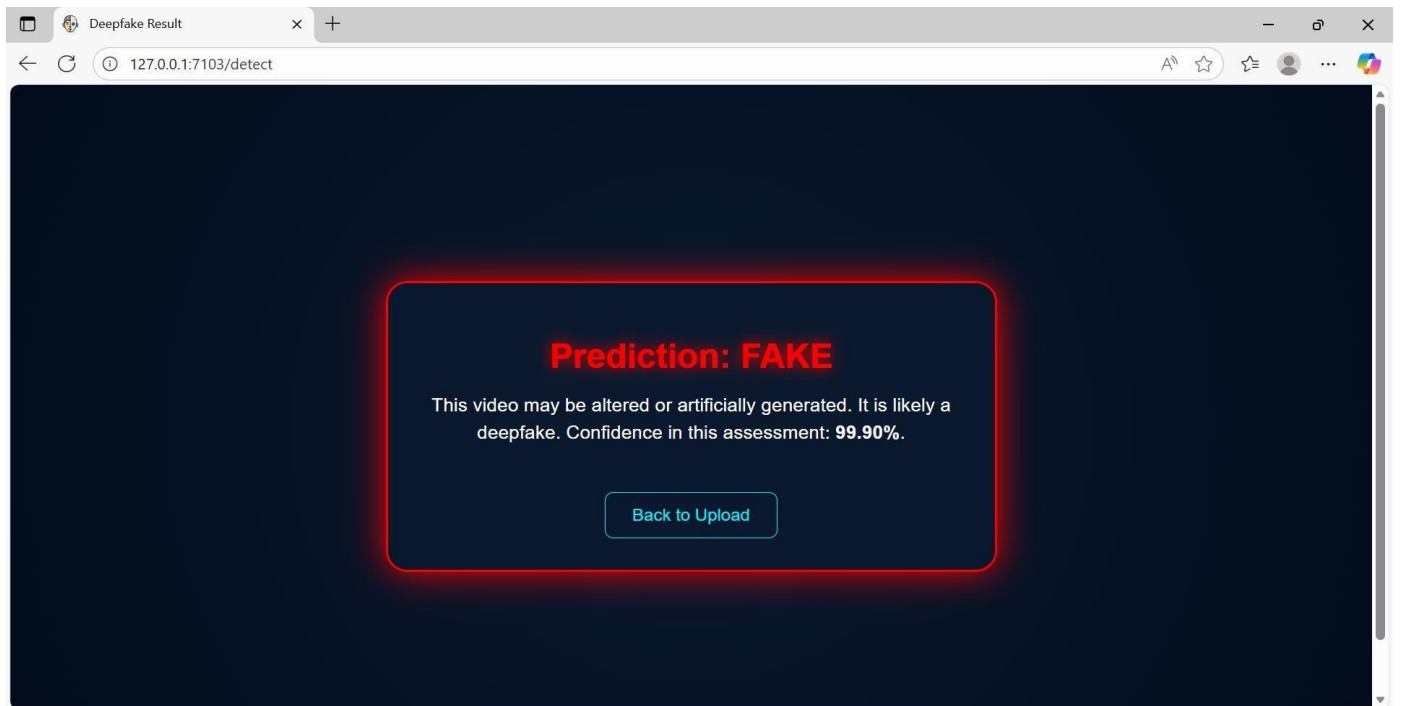


Figure 8.4. Deepfake Detection Result page

CHAPTER 9

RESULTS AND DISCUSSION

The experimental evaluation of the proposed Deepfake video detection framework demonstrates that integrating advanced spatial and temporal models significantly enhances detection performance. Models trained on raw video frames achieved an accuracy of 70–85%, indicating limited capability in capturing subtle temporal manipulation artifacts. Incorporating face extraction and CNN-based feature learning improved accuracy to 82–92% by isolating manipulated facial regions and reducing background noise.

Further improvements were observed with the addition of Efficient Net and Bi-LSTM, which achieved an accuracy of 90–96%, leveraging effective spatial–temporal feature representations. The integration of Vision Transformer (ViT) further enhanced spatial feature extraction by capturing global contextual relationships across frames, improving the model’s ability to detect subtle manipulations that may be missed by convolutional architectures. The combination of Efficient Net, ViT, and Bi-LSTM with an attention mechanism achieved the highest accuracy, ranging from 96–99%, with a best validation accuracy of 95.08%.

These results indicate that single-frame analysis alone is insufficient for robust Deepfake detection, as temporal inconsistencies across frames are often subtle. Face extraction improves robustness by focusing on critical facial regions, while Bi-LSTM captures sequential dependencies between frames. The ViT component enhances global spatial understanding, enabling detection of manipulations across larger facial regions. Although minor performance degradation occurs in highly compressed or low-resolution videos, the proposed framework demonstrates strong generalization across multiple datasets, confirming its suitability for real-world Deepfake detection applications.

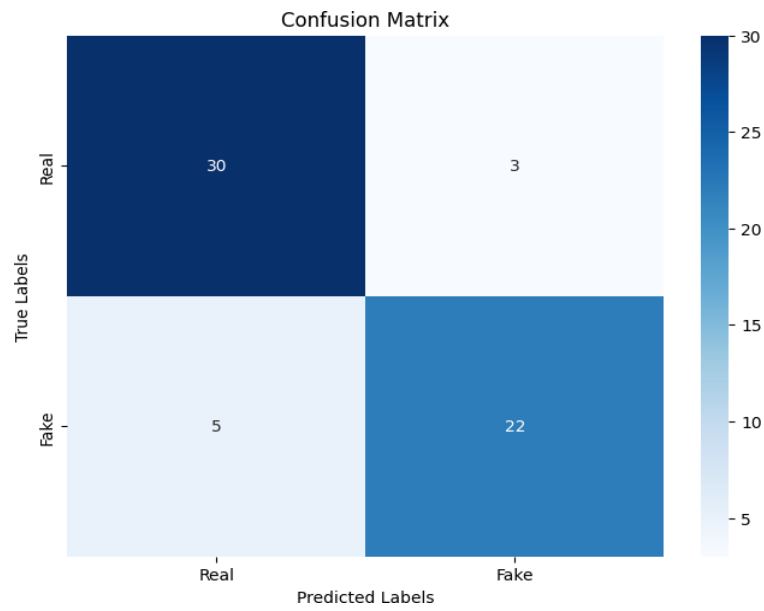


Figure 9.1. Confusion Matrix Representation

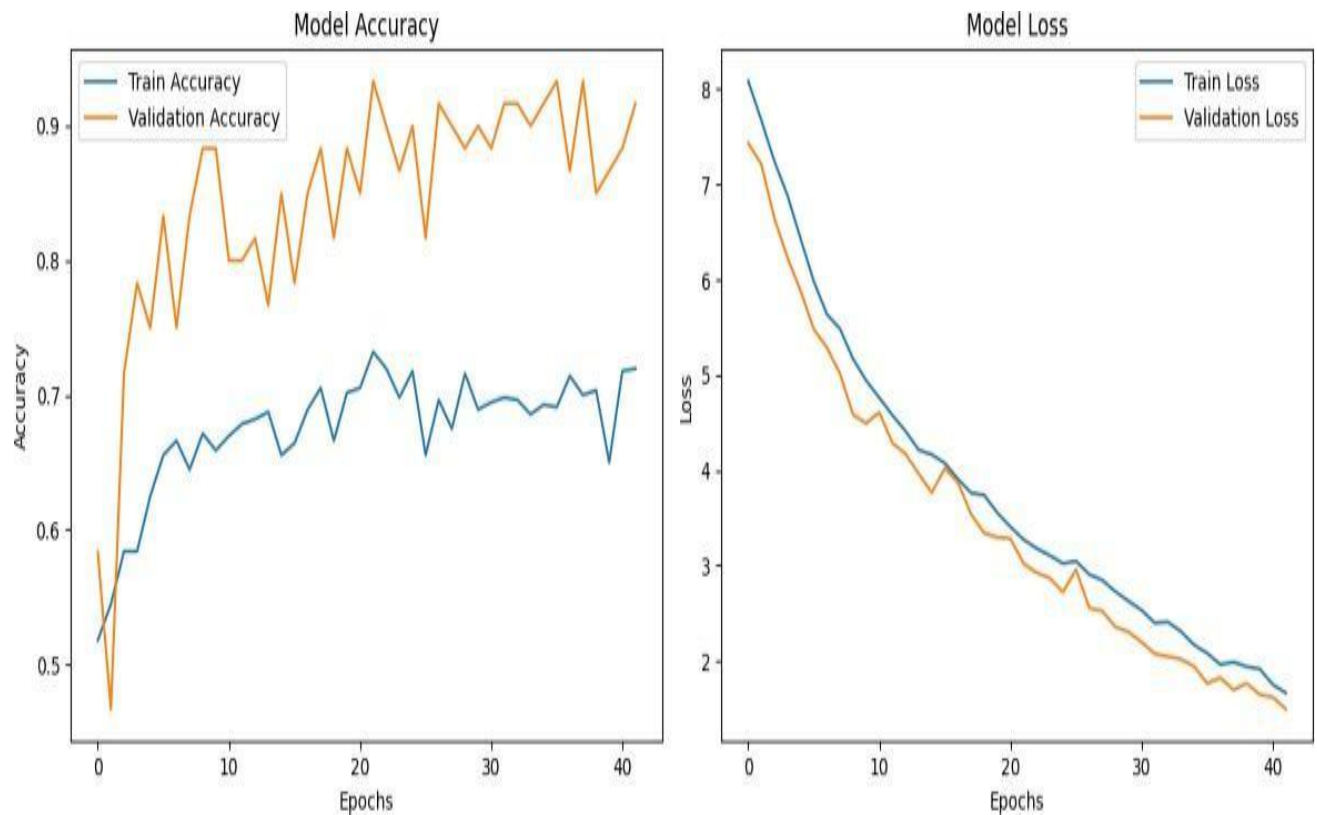


Figure 9.2. Training and Validation Accuracy Graph

```

.. 2/2 ----- 31s 15s/step
Test Accuracy: 86.67%
Classification Report:
      precision    recall  f1-score   support

     REAL       0.86      0.91      0.88        33
     FAKE       0.88      0.81      0.85        27

 accuracy                   0.87        60
 macro avg       0.87      0.86      0.86        60
 weighted avg    0.87      0.87      0.87        60

Epoch 40 | Train 0.0181/99.18% | Val 0.2211/93.44%
Epoch 41 | Train 0.0215/99.39% | Val 0.2299/94.26%
Epoch 42 | Train 0.0188/99.39% | Val 0.2249/94.26%
Epoch 43 | Train 0.0177/99.18% | Val 0.2306/93.44%
Epoch 44 | Train 0.0185/99.59% | Val 0.2493/93.44%
Epoch 45 | Train 0.0179/99.59% | Val 0.2769/93.44%
Epoch 46 | Train 0.0292/98.77% | Val 0.2310/94.26%
Epoch 47 | Train 0.0598/98.57% | Val 0.2503/94.26%
Epoch 48 | Train 0.0407/98.57% | Val 0.2828/94.26%
Epoch 49 | Train 0.0444/97.75% | Val 0.2386/94.26%
Epoch 50 | Train 0.0222/99.18% | Val 0.2315/93.44%
Epoch 51 | Train 0.0284/98.98% | Val 0.2243/94.26%
Epoch 52 | Train 0.0216/99.18% | Val 0.3325/91.80%
Epoch 53 | Train 0.1037/96.31% | Val 0.4685/90.98%
Epoch 54 | Train 0.0914/97.34% | Val 0.2338/94.26%
[Epoch 55] New best val_acc=95.08%, saved!
Epoch 55 | Train 0.0684/97.34% | Val 0.2050/95.08%
Epoch 56 | Train 0.0857/96.93% | Val 0.1801/93.44%
Epoch 57 | Train 0.0491/98.16% | Val 0.2235/95.08%
Epoch 58 | Train 0.0398/98.36% | Val 0.2710/93.44%
Epoch 59 | Train 0.0262/98.77% | Val 0.2314/94.26%
Epoch 60 | Train 0.0264/98.98% | Val 0.2980/94.26%
Best Val Acc: 95.08196721311475

```

Figure 9.3 Accuracy, Precision, Recall and F1-Score Comparison

CONCLUSION AND FUTURE ENHANCEMENT

CONCLUSION

The Deepfake Video Detection System developed in this project demonstrates high accuracy, robustness, and practical applicability in detecting manipulated videos. The model, based on EfficientNetB7, Vision Transformer (ViT), combined with Bi-LSTM and an attention mechanism, achieved a training accuracy of up to 99.39% and a validation accuracy of 95.08%. This indicates that the system effectively learns both local spatial features (via Efficient Net), global contextual features (via ViT), and temporal inconsistencies (via Bi-LSTM), while maintaining strong generalization to unseen data and minimizing overfitting.

The system was also successfully tested on YouTube videos, accurately detecting Deepfakes and demonstrating its practical applicability for real-world online content monitoring. Visualization tools, including confusion matrices, ROC curves, and attention heatmaps, further validate the model's reliability, interpretability, and its ability to emphasize critical regions prone to manipulation. These results confirm that the proposed framework is highly suitable for applications in digital forensics, social media monitoring, and cybersecurity, offering a comprehensive solution for combating the growing threat of Deepfake content.

FUTURE ENHANCEMENT

Multimodal Analysis Integration: Incorporate audio cues, facial micro-expressions, and speech patterns alongside visual features to detect cross-modal inconsistencies and significantly boost detection accuracy. **Advanced Attention and Transformer Mechanisms:** Integrate self-attention or Vision Transformer (ViT) modules to dynamically focus on critical regions and capture fine-grained spatial-temporal inconsistencies in manipulated videos. **Real-Time Streaming and Social Media Integration:** Extend support to live streaming platforms and social media APIs for continuous, real-time monitoring and proactive deepfake detection. **Federated Learning for Privacy-Preserving Updates:** Implement federated learning to enable

decentralized model improvement using user-contributed data while ensuring data privacy and compliance with regulations.

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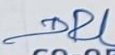
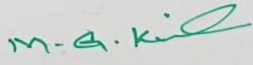
CERTIFICATES

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INTERNATIONAL CONFERENCE ON ENGINEERING, SCIENCE AND TECHNOLOGY
(ICEST-2K26)

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This is to certify that Dr./Mr./Ms. ADHINARAYANAN S of
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FOR STRENGTHENING SOCIAL MEDIA SECURITY 2
International Conference on Engineering, Science and Technology
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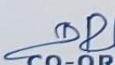
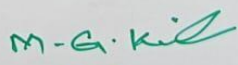
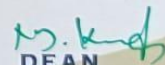
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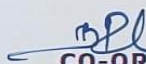
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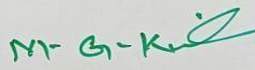
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This is to certify that Dr./Mr./Ms. SURIYA MOORTHY K of
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